

Binary Operation

A binary operation 'o' on a non-empty set A is a mapping which associates with each ordered pair (a, b) of elements of A, a uniquely defined element $c \in A$. Thus 'o' is a mapping of the product set $A \times A$ to A.

Symbolically, a map $o: A \times A \rightarrow A$ is called binary operation on set A.

- * If $(a, b) \in A \times A$ then o-image of (a, b) is denoted by aob
- * If the set A is closed w.r.t. the composition o, then we say that o is a binary operation on A. (Closure property)
- * We also use other symbols like $*$, $+$, $\oplus$, $\wedge$, $\vee$ etc. for binary operations.

Ex: i) Addition is a binary operation on the set of integers.

(ii) Multiplication is a binary operation on the set of natural numbers.

(iii) Subtraction on $\mathbb{I}$?, on $\mathbb{N}$?

(iv) Division on $\mathbb{I}$?

* Also known as binary composition.

* If ' $*$ ' is a binary composition on a non-empty set G then $a * b \in G \quad \forall a, b \in G$.

Algebraic Structure

A non-empty set G equipped with one or more binary operation(s) is called an algebraic structure. If $*$ is a binary operation on G then we denote algebraic structure as $(G, *)$.

* Also known as Groupoid or quasi group.

Ex: $(\mathbb{N}, +)$, $(\mathbb{I}, +)$, $(\mathbb{I}, -)$, $(\mathbb{R}, +, \cdot)$ are all algebraic structures.

Ex: $(\mathbb{N}, -)$ is not an algebraic structure.

Group

Let G be a non-empty set equipped with a binary operation denoted by $*$, then the algebraic structure $(G, *)$ is called as Group if the following conditions hold:

- (1) Closure Property: $a * b \in G \Rightarrow \forall a, b \in G$
- (2) Associativity: $(a * b) * c = a * (b * c) \quad \forall a, b, c \in G$
- (3) Existence of identity element: There exists an element $e \in G$ s.t.

$$a * e = a = e * a \quad \forall a \in G$$

This element e is known as identity element.

- (4) Existence of inverse: - For each element $a \in G$ there exists an element $b \in G$ s.t.

$$a * b = e = b * a$$

The element b is called as inverse of a i.e. $b = a^{-1}$.

* Set satisfying first two conditions is called 'semi group' and set satisfying condⁿs (1), (2) & (3) is known as 'monoid'.

Ex $(\mathbb{I}, +)$ is a group
or set of integers is a group under addition

Solⁿ Set of integers forms a group under addition if it satisfies the following conditions:

(i) Closure Property: We know that for any integers $a, b \in \mathbb{I}$

$$a + b \in \mathbb{I}$$

i.e. $a + b$ is again an integer

So we can say $\mathbb{I}$ is closed w.r.t. addition.

(ii) Associativity: For any three integers

$$a, b, c \in \mathbb{I},$$

$$(a + b) + c = a + (b + c)$$

(iii) Existence of identity element: For any $a \in \mathbb{I}$, $\exists 0 \in \mathbb{I}$, s.t.

$$a + 0 = 0 + a = a$$

0 is the additive identity in the set of integers

(iv) Existence of Inverse: We know that for any $a \in \mathbb{I}$, $\exists -a \in \mathbb{I}$ s.t.

$$a + (-a) = 0 = (-a) + a$$

So $-a = a^{-1}$ for each a in $\mathbb{I}$
 $\Rightarrow$ additive inverse of each element exists in $\mathbb{I}$

Since $(\mathbb{I}, +)$ satisfies all the conditions
Hence $(\mathbb{I}, +)$ is a group or $\mathbb{I}$ is a group under addition.

~~Ex~~ $(\mathbb{N}, +)$ is not a group. (It's a semigroup)

~~Ex~~ $(\mathbb{I}^+, +)$ is not a group.

~~Ex~~ $(\mathbb{I}, -)$ isn't a group.

~~Ex~~ Does $(\mathbb{I}, \cdot)$ form a group? where ' $\cdot$ ' is usual multiplication.

Abelian Group or Commutative group

A group G is said to be abelian or commutative if in addition to the previous four conditions the following condition is also satisfied:

(5) Commutativity: $a * b = b * a \quad \forall a, b \in G$

~~Ex~~ $(\mathbb{I}, +)$ is an abelian group

Finite and Infinite Group (Order of the Group)

If in a group G the underlying set G consists of a finite number of distinct elements then the group is called 'finite group' otherwise an 'infinite group'.

* The no. of the elements (distinct) in a finite group is called the 'order' of the group. An infinite group is said to be of 'infinite order'.

* Order of a group G is denoted by $o(G)$.

Some General Properties of Groups

- * Uniqueness of identity i.e. the identity element 'e' in a group is unique.
- * Uniqueness of inverse i.e. the inverse of each element of a group is unique.
- * The identity element e is its own inverse i.e. $e^{-1} = e$.
- * If inverse of a is a^{-1} then inverse of a^{-1} is a i.e. $(a^{-1})^{-1} = a$.
- * $(ab)^{-1} = b^{-1}a^{-1} \quad \forall a, b \in G$.
It is known as 'reversal rule'. It can be generalised as
If $a, b, c, \dots, j, k \in G$ then
 $(abc \dots jk)^{-1} = k^{-1}j^{-1} \dots c^{-1}b^{-1}a^{-1}$
- * Cancellation laws: If $a, b, c \in G$, then
 $ab = ac \Rightarrow b = c$ (left cancellation law)
 $ba = ca \Rightarrow b = c$ (right cancellation law)

Composition table for finite sets

A binary composition in a finite set can be shown in a tabular form known as 'Composition table'.

* The composition table for a finite group contains each element exactly once in each of its rows & columns.

Ex: Show that the four fourth roots of unity namely $\{1, -1, i, -i\}$ form a group w.r.t. multiplication.

Sol: Let $G = \{1, -1, i, -i\}$. To show that multiplication is a composition in G , we form the composition table:

$\times$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

Now we make the following conclusions:

- (1) Closure Property: Since all entries in the composition table are elements of the set G , G is closed w.r.t. multiplication.
- (2) Associativity: The elements of G are complex numbers and the multiplication of complex numbers is associative.
- (3) Existence of identity element: From the composition table we see that the row headed by the element 1 just coincides with top row of the composition table indicating all the elements of G .

i.e. $1 \in G$ and $1 \cdot a = a \quad \forall a \in G$.
 $\Rightarrow 1$ is the identity element in G .

(4.) Existence of inverse: We know that inverse of identity element is the element itself. Also from the composition table we can see that

$$(-1)(-1) = 1 \quad (\text{identity element})$$

$$(-i)(i) = -i^2 = 1$$

$$(i)(-i) = -i^2 = 1$$

i.e. inverse of -1 is $-1 \in G$

inverse of $-i$ is $i \in G$

and inverse of i is $-i \in G$

$$\Rightarrow a^{-1} \in G \quad \forall a \in G$$

Hence $G = \{1, -1, i, -i\}$ is a group under multiplication.

* The multiplication of complex numbers is commutative, therefore G is an abelian group of order 4 w.r.t multiplication.

* To see that the composition is commutative we observe from the composition table that the entries in the rows just coincide with the corresponding entries in the corresponding columns.

Addition modulo m (a different type of addition)

It is written as $a +_m b$ where a and b are any integers and m is a fixed integer s.t.

$a +_m b = r$, $0 \leq r < m$
 where r is the least non-negative remainder when $(a+b)$ is divided by m .

Ex: $15 +_5 7 = 2$
 $9 +_{12} 3 = 0$
 $3 +_3 1 = 1$

Multiplication modulo m

written as $a \times_m b$ where a, b are any integers and m is a fixed integer s.t.

$a \times_m b = r$, $0 \leq r < m$
 where r is the least non-negative remainder when $a \cdot b$ is divided by m .

Ex: $8 \times_5 3 = 4$
 $5 \times_3 4 = 2$

- * The set $G = \{0, 1, 2, \dots, m-1\}$ of first m non-negative integers is a group under the operation 'addition modulo m '.
- * The set G of $(p-1)$ integers $\{1, 2, 3, \dots, p-1\}$, p being a prime number, is a finite abelian gp. of order $(p-1)$ under the composition 'multiplication modulo p '.

Ex: The set $G = \{0, 1, 2, 3, 4, 5\}$ is a finite abelian gp. of order 6 w.r.t. addition modulo 6.

Ex: The set $G = \{1, 2, 3, 4, 5, 6\}$ is a finite abelian gp. of order 6 w.r.t. multiplication modulo 7.

Composition table for $(G, +_6)$:

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Ex: Prove that $Z_{10} = \{1, 3, 7, 9\}$ is an abelian group w.r.t. multiplication modulo 10.

Order of an element

Let G be a group and $a \in G$, then the order of an element a is the least positive integer n such that $a^n = e$ (identity element)

* If there exists no such n , then the order of a is infinity or zero.

* If G is a finite group then $o(a) \leq o(G) \quad \forall a \in G$.

* $o(a^m) \leq o(a)$

* $o(a^{-1}) = o(a)$